Power Budget

Team Number:	308
Project Name:	
Team Member Names:	**Adrian,Jacob,Sam B, Sam M, Mo**
Version:	

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,							
All Major Components	**Component Name**	**Part Number**	**yVoltageRan**	**#**	**aximumCurre**	**lCurrent(mA)**	**Unit**
	ESP32 Module	ESP32-WROOM-	3.0V - 3.6V	1	1000	1000	mA
	Photoresistor	NORPS-12	3.3V - 5V	1	Passive	0	mA
						0	mA
						0	mA
						0	mA
						0	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in							
+12V Power Rail	**Component Name**	**Part Number**	**yVoltageRan**	**#**	**aximumCurre**	**lCurrent(mA)**	**Unit**
	Solenoid	(full part numbe	+12 - 24V	1	500	500	mA
	Stepper motor	(full part numbe	+12 - 24V	1	300	300	mA
						0	mA
						0	mA
						0	mA
					Subtotal	800	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	1000	mA
c1. Regulator or Source Ch	+12V regulator	LM7812	+12V - 35V	1	1000	1000	mA
					Total Remaining Current Available on +12V Rail	0	mA
+5V Power Rail	**Component Name**	**Part Number**	**yVoltageRan**	**#**	**aximumCurre**	**lCurrent(mA)**	**Unit**
	PSoC™ 4 BLE Module	(full part numbe	(range)	2	200	400	mA
	Opamp	(full part numbe	(range)	1	100	100	mA

						0	mA
						0	mA
						0	mA
					Subtotal	500	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	625	mA
c2. Regulator or Source Ch	+5V Regulator	LM7805	(range)	1	1000	1000	mA
					Total Remaining Current Available on +5V Rail	375	mA
-5V Power Rail	**Component Name**	**Part Number**	**yVoltageRan**	**#**	**aximumCurre**	**lCurrent(mA)**	**Unit**
	Opamp	(full part numbe	(range)	1	100	100	mA
						0	mA
						0	mA
						0	mA
					Subtotal	100	mA
					Safety Margin	25%	
					Total Current Required on -5V Rail	125	mA
c3. Regulator or Source Ch	-5V Regulator	(full part numbe	(range)	1	500	500	mA
					Total Remaining Current Available on -5V Rail	375	mA
+3.3V Power Rail	**Component Name**	**Part Number**	**yVoltageRan**	**#**	**aximumCurre**	**lCurrent(mA)**	**Unit**
	ESP32 Module	ESP32-WROOM-	3.0V - 3.6V	1	1000	1000	mA
						0	mA
						0	mA
						0	mA
					Subtotal	1000	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	1250	mA
c4. Regulator or Source Ch	+3.3V low-dropout regula	LM1085ISX-3.3	+5V - 15V	1	3000	3000	mA
					Total Remaining Current Available on 3.3V Rail	1750	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of							
External Power Source 1	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu**	**aximumCurre**	**lCurrent(mA)**	**Unit**
Power Source 1 Selection	AC-DC Wall Supply	Model 0930	110-220VAC	+9V	3000	3000	mA
						0	mA
Power Rails Connected to External Power Source 1						0	mA
	+3.3V low-dropout regulat	LM1085ISX-3.3	+5V - 15V	1	3000	3000	mA
	Total Remaining Current Available on External Power Source 1					0	mA
External Power Source 2	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu**	**aximumCurre**	**lCurrent(mA)**	**Unit**
Power Source 2 Selection	Battery	full part number	+9V	-9V	500	500	mA
	-5V Regulator	(full part numbe	(range)	1	500	500	mA
Power Rails Connected to External Power Source 2							
	Total Remaining Current Available on External Power Source 2					0	mA
E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the							
	Component Name	**Part Number**	**yVoltageRange**		**Capacity(mAh)**	**redByRegulators**	
	Battery	(full part numbe	+12V		500	500	
					Battery Life	1	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit